

DATA WAREHOUSE

Movie Rental Data Warehouse Design

By:

mustafa mahmoud

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Section 1: Introduction

This report presents a high-level data warehouse design for a movie rental business. The source system is an OLTP (Online Transaction Processing) database called Sakila, which supports daily operations such as recording rentals, processing payments, managing customers, films, inventory, staff, and store locations.

An OLTP database is designed for fast read and write operations on individual transactions. It is highly normalized meaning data is split across many tables to reduce redundancy which makes it efficient for daily operations but slow and complex for analytical reporting. For example, to find out which films generated the most revenue last year, a user would need to join six or more tables in the OLTP system.

A data warehouse is designed specifically for analytical reporting. It uses a dimensional model (star schema) that organizes data around business processes and makes it easy to answer questions like which films generated the most revenue this year, or which store had the highest rental activity last quarter.

The goal of this assignment is to study the Sakila OLTP schema and redesign it as a data warehouse using dimensional modeling principles, including the definition of fact tables, dimension tables, and an ETL process to move data from the operational system into the warehouse.

Feature	OLTP (Sakila)	Data Warehouse
Purpose	Daily transactions	Analytical reporting
Design	Normalized (many tables)	Dimensional (star schema)
Query type	Simple, fast inserts/updates	Complex, aggregated queries
Users	Staff, applications	Managers, analysts
Data	Current data only	Historical data over time
Example query	Insert a new rental	Revenue by month for last 3 years

Section 2: Business Questions

Before designing the data warehouse, we identify the analytical questions that business managers need to answer. These questions guide all design decisions about which fact tables, dimension tables, and measures to include.

#	Business Question	Why It Matters
1	Which films are rented most frequently?	Identifies popular inventory to stock more of
2	Which films generate the highest revenue?	Drives pricing and stocking decisions
3	Which film categories are most popular?	Guides purchasing of new films
4	Which stores have the highest rental volume?	Measures store performance
5	Which stores generate the highest revenue?	Supports budget and staffing decisions
6	Which customers rent the most films?	Identifies loyal customers for promotions
7	Which customers generate the highest revenue?	Supports customer segmentation
8	How does rental activity change by month/quarter/year?	Reveals seasonal trends
9	How does revenue change over time?	Tracks business growth
10	Which staff members process the most rentals?	Measures staff performance
11	Which cities or countries have the most active customers?	Supports location-based marketing
12	What is the average rental duration per film or category?	Helps set rental period policies
13	Which films are returned late most often?	Identifies late-return patterns
14	How does store performance differ by location?	Benchmarks stores against each other
15	What are the peak rental days or months?	Helps with staff scheduling
16	Which films have the most copies per store?	Helps with inventory planning
17	Which films run low on stock due to high demand?	Identifies need to purchase more copies
18	Which stores have underutilized inventory?	Supports stock redistribution decisions

Section 3: Dimensional Model Design

3.1 Selected Business Processes

Three main business processes exist in the Sakila OLTP system that are suitable for dimensional modeling:

- **Film Rental** a customer rents a film from a store, handled by a staff member. This is the core transaction of the business.
- **Payment** a customer makes a payment for a rental, processed by a staff member. This captures all revenue data.
- **Inventory Availability** tracks how many copies of each film exist per store and how often they are rented.

Each of these business processes becomes one fact table in the data warehouse.

3.2 Fact Tables

Fact Table 1: fact_rental

Property	Detail
Business process	A customer rents a film from a store
Grain	One row per rental transaction
Source OLTP tables	rental, inventory

Measure	Description
rental_duration_days	Actual number of days the film was kept
expected_duration_days	The allowed rental period defined for that film
late_return_flag	1 if returned late, 0 if returned on time
rental_count	Always 1 used for counting total rentals

Foreign Key	Links to
date_key	dim_date
customer_key	dim_customer
film_key	dim_film
store_key	dim_store
staff_key	dim_staff

Fact Table 2: fact_payment

Property	Detail
Business process	A customer makes a payment for a rental
Grain	One row per payment transaction
Source OLTP tables	payment

Measure	Description
amount	The payment amount in dollars
payment_count	Always 1 used for counting total payments

Foreign Key	Links to
date_key	dim_date
customer_key	dim_customer
staff_key	dim_staff
rental_key	fact_rental

Fact Table 3: fact_inventory

Property	Detail
Business process	Tracks how many copies of each film exist per store
Grain	One row per film per store
Source OLTP tables	inventory, rental

Measure	Description
total_copies	How many physical copies of this film exist in this store
times_rented	How many times this film has been rented from this store
availability_rate	Percentage representing how available the film is

Foreign Key	Links to
film_key	dim_film
store_key	dim_store

Summary of All Fact Tables

Fact Table	Business Process	Grain	Key Measures
fact_rental	Customer rents a film	One row per rental	rental_duration, late_return_flag

Fact Table	Business Process	Grain	Key Measures
fact_payment	Customer makes a payment	One row per payment	amount
fact_inventory	Film copies per store	One row per film per store	total_copies, times_rented, availability_rate

3.3 Dimension Tables

dim_date

Attribute	Description
date_key (PK)	Integer surrogate key in format YYYYMMDD
full_date	Actual date value
day_of_week	Monday, Tuesday...
day_number	Day of month: 1 to 31
month_number	Month number: 1 to 12
month_name	January, February...
quarter	1, 2, 3, or 4
year	2005, 2006, 2007
is_weekend	1 if Saturday or Sunday, 0 otherwise

Source: Generated programmatically in Python using pandas date_range

Purpose: Allows all fact tables to be analyzed by any time dimension

dim_customer

Attribute	Description
customer_key (PK)	Surrogate key
customer_id	Original OLTP customer ID
full_name	Combination of first_name and last_name
email	Customer email address
city	City where customer lives
country	Country where customer lives
active_status	Active or Inactive

Source OLTP tables: customer, address, city, country

Purpose: Allows rental and payment analysis by customer name, location, and status

dim_film

Attribute	Description
film_key (PK)	Surrogate key
film_id	Original OLTP film ID
title	Film title
description	Film description

Attribute	Description
release_year	Year the film was released
rating	G, PG, PG-13, R, NC-17
rental_duration	Number of days allowed for rental
rental_rate	Price per rental in dollars
length_minutes	Film length in minutes
language	Film language name
category	Film category name

Source OLTP tables: film, language, film_category, category

Purpose: Allows rental and inventory analysis by title, category, rating, and language

dim_store

Attribute	Description
store_key (PK)	Surrogate key
store_id	Original OLTP store ID
city	City where the store is located
country	Country where the store is located
address	Street address of the store
manager_name	Full name of the store manager

Source OLTP tables: store, address, city, country, staff

Purpose: Allows rental and payment analysis by store location and management

dim_staff

Attribute	Description
staff_key (PK)	Surrogate key
staff_id	Original OLTP staff ID
full_name	Combination of first_name and last_name
email	Staff email address
store_id	The store this staff member works at

Source OLTP tables: staff

Purpose: Allows rental and payment analysis by staff member to measure performance

dim_location

Attribute	Description
location_key (PK)	Surrogate key
address	Street address
city	City name
country	Country name
postal_code	Postal code

Source OLTP tables: address, city, country

Purpose: Allows geographic analysis of customers and stores

Summary of All Dimension Tables

Dimension	Main Attributes	Source OLTP Tables	Shared With
dim_date	date, day, month, quarter, year	Generated in Python	fact_rental, fact_payment
dim_customer	name, email, city, country	customer, address, city, country	fact_rental, fact_payment
dim_film	title, category, rating, language	film, language, film_category, category	fact_rental, fact_inventory
dim_store	city, country, address, manager	store, address, city, country, staff	fact_rental, fact_inventory
dim_staff	name, email, store	staff	fact_rental, fact_payment
dim_location	address, city, country	address, city, country	dim_customer, dim_store

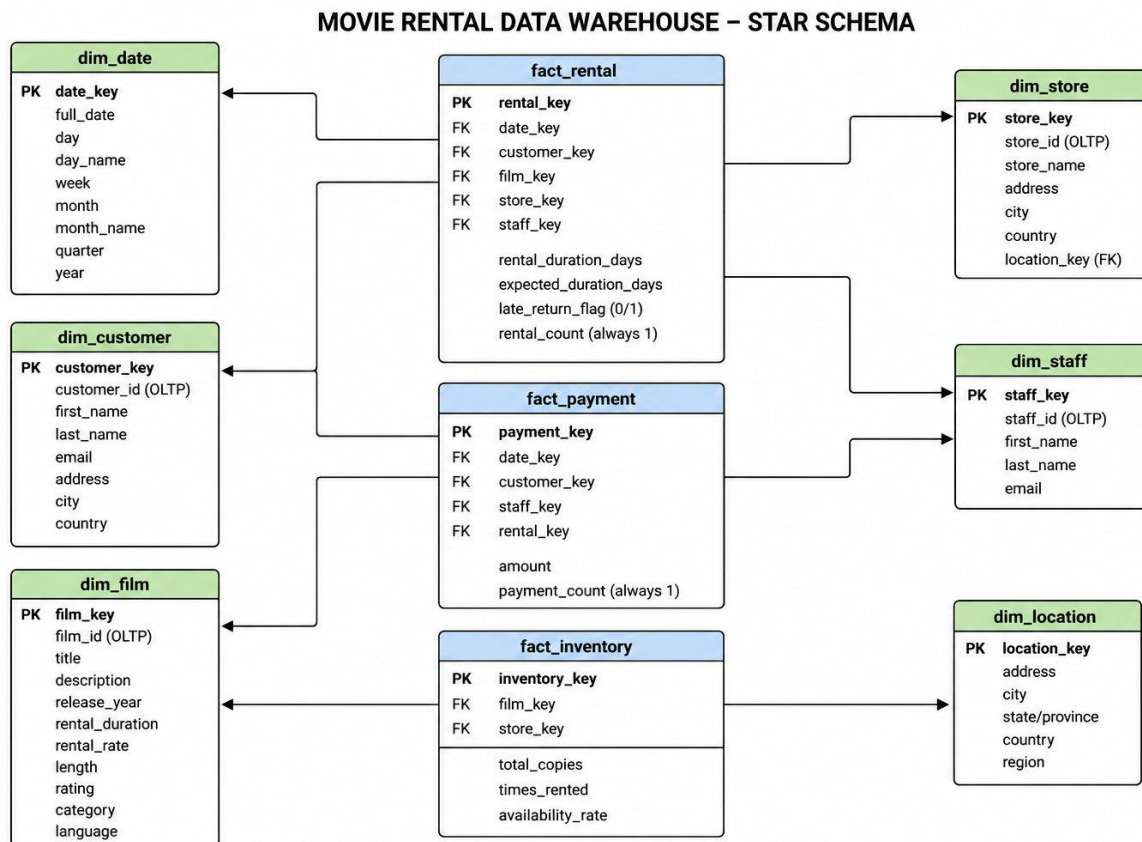
This design uses a star schema. Each fact table sits at the center and is directly connected to its dimension tables. Dimension tables are intentionally denormalized for example, dim_customer already contains city and country directly. This makes queries simpler and faster for analytical reporting.

Section 4: Dimensional Model Diagram

The proposed data warehouse uses a star schema design with three central fact tables surrounded by six dimension tables:

- fact_rental connected to: dim_date, dim_customer, dim_film, dim_store, dim_staff
- fact_payment connected to: dim_date, dim_customer, dim_staff, fact_rental
- fact_inventory connected to: dim_film, dim_store

The following diagram shows the complete star schema for the Movie Rental Data Warehouse, including all fact tables, dimension tables, primary keys, foreign keys, and relationships:



Schema Element	Type	Connected To
fact_rental	Fact Table (center)	dim_date, dim_customer, dim_film, dim_store, dim_staff
fact_payment	Fact Table (center)	dim_date, dim_customer, dim_staff, fact_rental
fact_inventory	Fact Table (center)	dim_film, dim_store
dim_date	Dimension Table	fact_rental, fact_payment
dim_customer	Dimension Table	fact_rental, fact_payment
dim_film	Dimension Table	fact_rental, fact_inventory
dim_store	Dimension Table	fact_rental, fact_inventory
dim_staff	Dimension Table	fact_rental, fact_payment
dim_location	Dimension Table	dim_customer, dim_store

Section 5: ETL Design

5.1 Extract

The following OLTP tables are extracted from the Sakila source database using pandas read_sql with a SQLAlchemy connection:

OLTP Table	Why It Is Needed
rental	Core rental transactions main source for fact_rental
payment	Payment transactions main source for fact_payment
inventory	Links rentals to specific film copies and stores
film	Film attributes for dim_film
customer	Customer attributes for dim_customer
store	Store attributes for dim_store
staff	Staff attributes for dim_staff and dim_store
address	Address details used in dim_customer, dim_store, dim_location
city	City names used in all location-based dimensions
country	Country names used in all location-based dimensions
category	Film category names used in dim_film
film_category	Links films to their categories
language	Film language names used in dim_film

5.2 Transform

The following transformations are applied before loading data into the warehouse:

- Combine name fields: Concatenate first_name and last_name into full_name for customers and staff.
- Join multiple OLTP tables using pandas merge: For example, dim_customer is built by merging customer + address + city + country into one flat denormalized row.
- Generate surrogate keys: Each dimension table uses AUTO_INCREMENT in MySQL. The database automatically assigns the next integer as the surrogate key when rows are inserted.
- Create date keys: Convert rental_date and payment_date into integer keys in format YYYYMMDD using strftime.
- Calculate rental duration: Compute the actual number of days kept using (return_date - rental_date).dt.days
- Detect late returns: Compare actual rental duration against the allowed rental_duration. Set late_return_flag = 1 if returned late.

- Handle NULL return dates: Rentals with no return date are excluded from fact_rental since the film has not been returned yet.
- Calculate availability rate: For fact_inventory, computed as $(1 - \text{times_rented} / (\text{total_copies} * 100)) * 100$, clipped at 0 minimum.
- Resolve surrogate keys: After dimensions are loaded, fact tables look up their surrogate keys using pandas merge on original OLTP IDs.

5.3 Load

Dimensions are always loaded before fact tables because fact tables contain foreign keys that reference dimension surrogate keys. If fact tables are loaded first, the foreign key references do not exist and the load will fail.

Step	Table	Reason
1	dim_date	No dependencies generated independently
2	dim_location	No dependencies built from address/city/country
3	dim_customer	Depends on address data
4	dim_film	Depends on language and category
5	dim_store	Depends on address and staff
6	dim_staff	Depends on staff table
7	fact_rental	Depends on all 6 dimensions above
8	fact_payment	Depends on dim_date, dim_customer, dim_staff, fact_rental
9	fact_inventory	Depends on dim_film and dim_store

All tables are loaded using pandas to_sql with if_exists='append'. Before each load, the table is cleared using DELETE to allow re-running the ETL without creating duplicates.

Section 6: Data Quality Considerations

#	Rule	Check	Action if Failed
1	No NULL rental dates	rental_date IS NOT NULL	Reject record, log the rental_id
2	No negative payment amounts	amount > 0	Reject record, log the payment_id
3	Return date after rental date	return_date >= rental_date	Flag as invalid, exclude from fact_rental
4	No duplicate rental IDs	COUNT(*) = COUNT(DISTINCT rental_id)	Deduplicate before loading
5	All film keys must exist in dim_film	FK constraint check before fact load	Log missing film IDs
6	All customer keys must exist in dim_customer	FK constraint check	Log orphan rental records
7	All date keys must exist in dim_date	LEFT JOIN dim_date WHERE date_key IS NULL	Extend dim_date date range
8	Payment count must match source	COUNT(fact_payment) = COUNT(sakila.payment)	Investigate and reload
9	Rental duration must not be negative	rental_duration_days >= 0	Flag as invalid, exclude
10	No NULL surrogate keys in fact tables	Check all FK columns for NULL after load	Trace ETL join logic and fix

Section 7: Conclusion

This report proposed a complete high-level data warehouse design for the Sakila movie rental OLTP system. The design transforms a normalized operational database into a dimensional model optimized for analytical reporting and business decision-making.

The proposed data warehouse includes three fact tables: `fact_rental` captures every film rental transaction with 15,861 records loaded, `fact_payment` captures every payment transaction, and `fact_inventory` tracks film copy availability per store with 1,521 records covering all film-store combinations. Together these three fact tables cover the three most important business processes in the movie rental operation.

Six dimension tables `dim_date`, `dim_customer`, `dim_film`, `dim_store`, `dim_staff`, and `dim_location` provide the descriptive context needed for filtering, grouping, and drilling down in analytical reports. The star schema design ensures that queries are simple and fast, even when analyzing millions of records across years of history.

The ETL process was implemented in Python using `pandas` and `SQLAlchemy`. It extracts data from 13 OLTP source tables, applies transformations including table joins using `pandas merge`, surrogate key generation, date key creation, late-return detection, name concatenation, and data cleaning, then loads the warehouse in the correct order dimensions first, fact tables second.

The 18 business questions identified in Section 2 can all be answered efficiently using this data warehouse design, covering rental activity, revenue trends, customer behavior, film popularity, store performance, staff effectiveness, and inventory management.

This data warehouse gives business managers a clean, consistent, fast, and historically rich analytical database that supports better decisions across every area of the movie rental business. The successful ETL execution confirmed that all data was loaded correctly, with the most rented film (BUCKET BROTHERHOOD with 34 rentals) being immediately queryable from the warehouse.